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RESEARCH ARTICLE - SPECIAL ISSUE - CHEMISTRY

Conformational Preferences and Electrochemical Performance of Ethyleneoxy Phenylboronate Electrolyte Additives

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Abstract It is generally accepted that the structural characteristics of a molecule determine its physical and electrochemical properties. In this study, the conformations and some electrochemical properties of various boronates were investigated through computational study using density functional theory (DFT) with the Becke's three-parameter hybrid method utilizing the Lee–Young–Parr correlation functional (B3LYP). After initial energy optimization using Møller–Plesset perturbation theory (MP2), the conformational preferences and energetics were investigated using DFT calculations and the 6-31G(d,p) basis set in vacuo. The calculations and first results show that the ethyleneoxyboronates can be expected to perform well as redox shuttles, and boron-based redox shuttles can contribute to overcharge protection and safer batteries. HOMO–LUMO energy differences also indicate higher reactivities of the boronates, contributing to better solid electrolyte interphase formation.

Keywords Boronate · Electrolyte additive · Lithium-ion battery · Conformation · LUMO energy

1 Introduction

As an alternative energy source, lithium-ion batteries (LIBs) have become increasingly important with a wide range of applications in industry, and many international companies are investing in this big project. The main aim of our work is to find new additives that will improve the safety, capacity and cycling behaviour of lithium-ion power sources [1–4], extending our previous work on siloxane [5–7] and boronic acid [8] additives.

Overcharge usually takes place when the current is forced through a cell, and the charge delivered is above its charge storing capability; this can lead to the chemical and electrochemical reaction of the battery's components, rapid temperature increase, self-accelerating reactions and finally explosion [9]. Overcharge protection additives may be classified as redox shuttles, which reversibly protect the cell from overcharging, or shutdown additives, which permanently terminate cell operation.

Boronates are known as redox shuttles and film-forming additives, and ethylene glycol oligomers have found wide application in lithium-ion battery electrolytes. For example, methyl phenyl bis-methoxydiethoxysilane can effectively prevent the decomposition and the co-intercalation of propylene carbonate (PC) [7]. A novel multifunctional electrolyte additive, 3,5-bis(trifluoromethyl) benzeneboronic acid (BA, Compound **7**, Scheme 1), is reduced at a higher potential than that of PC-solvated Li^+ ion because of its lower lowest unoccupied molecular orbital (LUMO) energy level [8]. This additive greatly suppresses the decomposition of PC and the exfoliation of graphite, allowing lithium ions to reversibly intercalate into and deintercalate from the graphite in PC-based electrolyte.

Based on these results, it was decided to combine these two functions by synthesizing novel ethylene oxide–boronate derivatives. Specifically, we set out to do theoretical (molecular modelling) studies of the HOMO and LUMO energies of these derivatives with respect to their expected electrochemical properties compared to those of the commonly used electrolyte solvents. We also synthesized novel ethyleneoxy (EO)-boronate derivatives, based on theoretical (molecular

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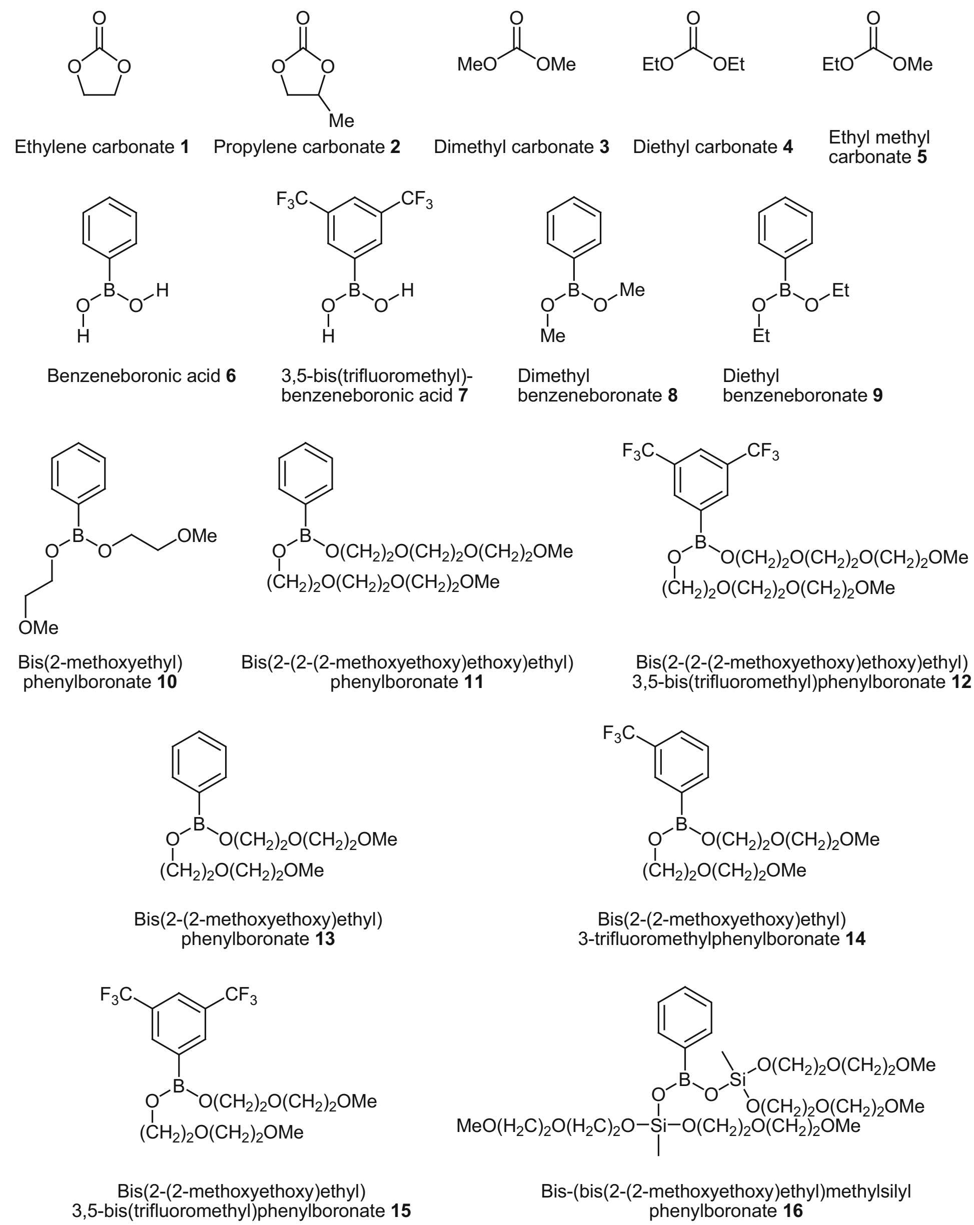
Scheme 1 Structures of Compounds **1** to **16**



modelling) studies of the HOMO and LUMO energies of these derivatives [8,10]. Here we report our first results.

2 Experimental

2.1 Computational Methods

In this study, the conformations and redox properties of various boronates were investigated through computational study [11] using density functional theory (DFT) with the Becke's three-parameter hybrid method utilizing the Lee–Young–Parr correlation functional (B3LYP). After initial energy optimization using Møller–Plesset perturbation theory (MP2), the conformational preferences and energetics were investigated using DFT calculations and the 6-31G(d,p) basis set in vacuo. Frequency calculations confirmed that energy minima were indeed found. Visualization was by means of Gaussview [12]. The compounds with lower LUMO energy than that of PC are assumed to decompose on the anode before the PC solvent because they are better electron acceptors than PC. The energy gap between each molecule's HOMO and LUMO was calculated as

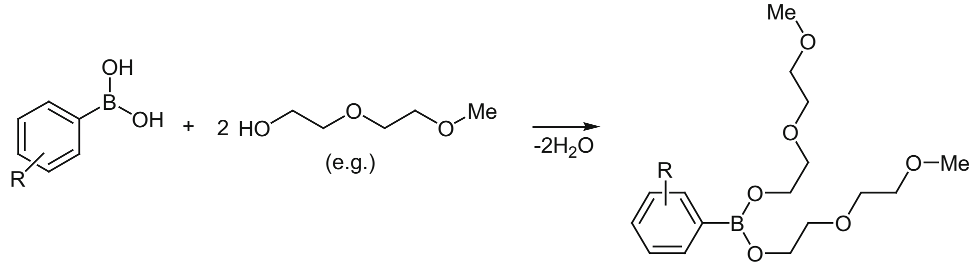
$$\Delta E = |E_{\mathrm{HOMO}} - E_{\mathrm{LUMO}}|$$

2.2 Chemistry

2.2.1 General Conditions

All reactions were monitored by thin-layer chromatography (TLC). Infrared (IR) spectra were recorded on a Bruker Alpha

Scheme 2 Synthesis of boronates **7** to **16**



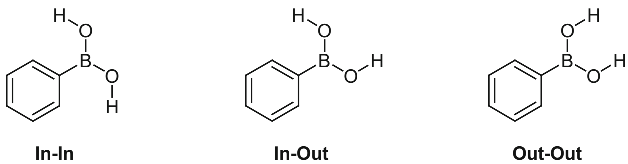


Fig. 1 Benzeneboronic acid geometries

FTIR spectrometer. [1]H and [13]C nuclear magnetic resonance (NMR) spectra were recorded using solvent as the internal standard with a Bruker Avance 400 spectrometer at 400 MHz and 100 MHz, respectively.

The arylboronates were prepared by reaction of the respective arylboronic acid with anhydrous diethylene or triethylene glycol monomethyl ether (Scheme 2) as follows: arylboronic acid (10.5 mmol) was dissolved in 100 ml anhydrous benzene, and a Dean-Stark trap was used to remove about 20 ml benzene–water azeotrope from the system to ensure anhydrous conditions. The solution was cooled down to about 70 °C, and 20 mmol diethylene or triethylene glycol monomethyl ether (EO) was added. The solution was heated under reflux until TLC indicated completion of the reaction (consumption of arylboronic acid). The solvent and excess EO were removed by vacuum distillation (240 °C at 1 mmHg) to yield a colourless oil.

2.2.2 Bis(2-(2-(2-methoxyethoxy)ethoxy)ethyl) phenylboronate 11

IR (neat): 3450, 3030, 2878, 1593, 1456 (asym. B–O), 1354 (asym. B–O), 1199, 1108, 1083 (sym. B–O), 967 cm^{-1}. ^{1}H-NMR ($CDCl_3$): δ_H 3.19 (6H, s, O–Me), 3.3–3.6 (16H, m, 8× $-CH_2-$), 7.20 (3H, m, arom.), 7.47 (2H, m, arom.).

2.2.3 Bis(2-(2-(2-methoxyethoxy)ethoxy)ethyl) 3,5-bis-(trifluoromethyl)phenylboronate 12

IR (neat): 2869, 1277, 1124 (sym. B–O), 710, 681 cm^{-1}. ^{1}H-NMR ($CDCl_3$, 400 MHz): δ_H 3.24 (6H, s, 2× –O*Me*), 3.3–3.5 (20H, m, 10× $-CH_2-$), 3.94 (4H, 2× $-CH_2-$), 7.62 (1H, s, arom.), 8.07 (1H, s, arom.). ^{13}C-NMR ($CDCl_3$, 100 MHz): δ_C 61.0 (2× –O*Me*), 66.2, 69.3, 70.1 ($-CH_2-$), 123.2 (–*C*H– arom.), 127.6 (arom.), 130.0 (arom.), 133.9 (–*C*H– arom.).

2.2.4 Bis(2-(2-methoxyethoxy)ethyl) phenylboronate 13

IR (neat): 3054, 2878, 1573, 1440 (asym. B–O), 1341 (asym. B–O), 1098, 1065 (sym. B–O), 1021, 702, 648 cm^{-1}. ^{1}H-NMR ($CDCl_3$, 300 MHz): δ_H 3.26 (6H, s, 2× OMe), 3.4–3.6 (12H, m, 6× $-CH_2-$), 4.09 (4H, s, 2× $-CH_2-$), 7.24 (3H, m, arom.), 7.70 (2H, m, arom.). ^{13}C-NMR ($CDCl_3$, 100 MHz): δ_C = 58.8 (2× –O*Me*), 70.1, 70.4, 70.5 ($-CH_2-$), 127.5 (–*C*H– arom.), 127.6 (arom.), 130.3 (arom.), 133.9 (–*C*H– arom.).

2.2.5 Bis(2-(2-methoxyethoxy)ethyl) 3-trifluoromethylphenylboronate 14

IR (neat): 2877, 1345 (asym. B–O), 1102, 1071 (sym. B–O), 707 cm^{-1}. ^{1}H-NMR ($CDCl_3$, 300 MHz): δ_H 3.25 (6H, s, 2× O–Me), 3.4–3.6 (12H, m, 6× $-CH_2-$), 4.07 (4H, m, 2× $-CH_2-$), 7.43 (3H, m, arom.), 7.80 (1H, m, arom.). ^{13}C-NMR ($CDCl_3$, 100 MHz): δ_C 30.5 ($-CF_3$), 58.7 (2× –O*Me*), 66.2, 69.3, 70.1 ($-CH_2-$), 123.2 (–*C*H– arom.), 129.3 (arom.), 129.6 (arom.), 133.9 (–*C*H– arom.).

2.2.6 Bis(2-(2-methoxyethoxy)ethyl) 3,5-bis(trifluoromethyl)phenylboronate 15

IR (neat): 2878, 1278, 1171, 1108 (sym. B–O), 843, 710, 681 cm^{-1}. ^{1}H-NMR ($CDCl_3$, 300 MHz): δ_H 3.46 (6H, s, 2× O–Me), 3.5–3.6 (12H, m, 6× $-CH_2-$), 4.22 (4H, m, 2× $-CH_2-$), 7.86-8.18 (3H, m, arom.). ^{13}C-NMR ($CDCl_3$, 100 MHz): δ_C 58.6 (2× –O*Me*), 66.3, 70.1, 71.8 ($-CH_2-$), 123.5 (–*C*H– arom.), 127.5 (arom.), 130.2 (arom.), 134.0 (–*C*H– arom.).

2.2.7 Bis-(bis(2-(2-methoxyethoxy)ethyl)methylsilyl) phenylboronate 16

IR (neat): 3450, 3030, 2878, 1593, 1456 (asym. B–O), 1354, 1199, 1108, 1083 (sym. B–O), 967 cm^{-1}. ^{1}H-NMR ($CDCl_3$): δ_H 0.23 (3H, s, Si–Me), 3.19 (6H, s, OMe), 3.3–3.6 (16H, m, 8×$-CH_2-$), 7.20 (3H, m, arom.), 7.47 (2H, m, arom.).

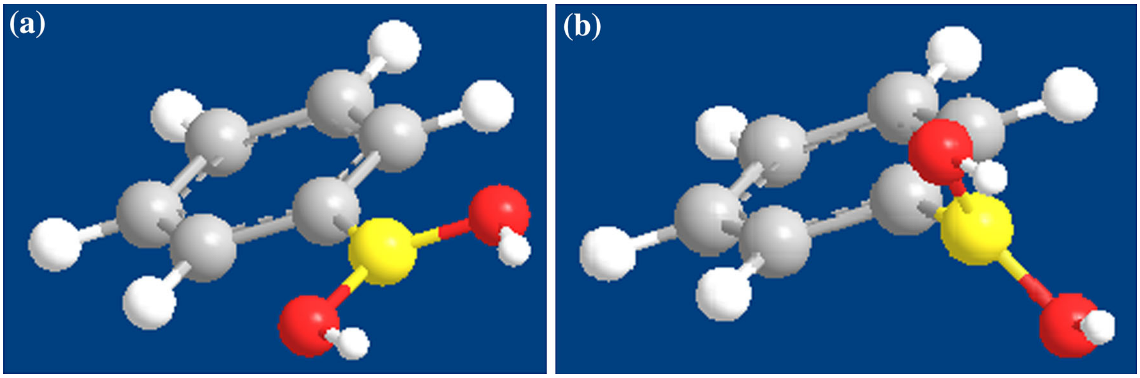


Fig. 2 Benzeneboronic acid dihedral angles: **a** 0° and **b** 90°

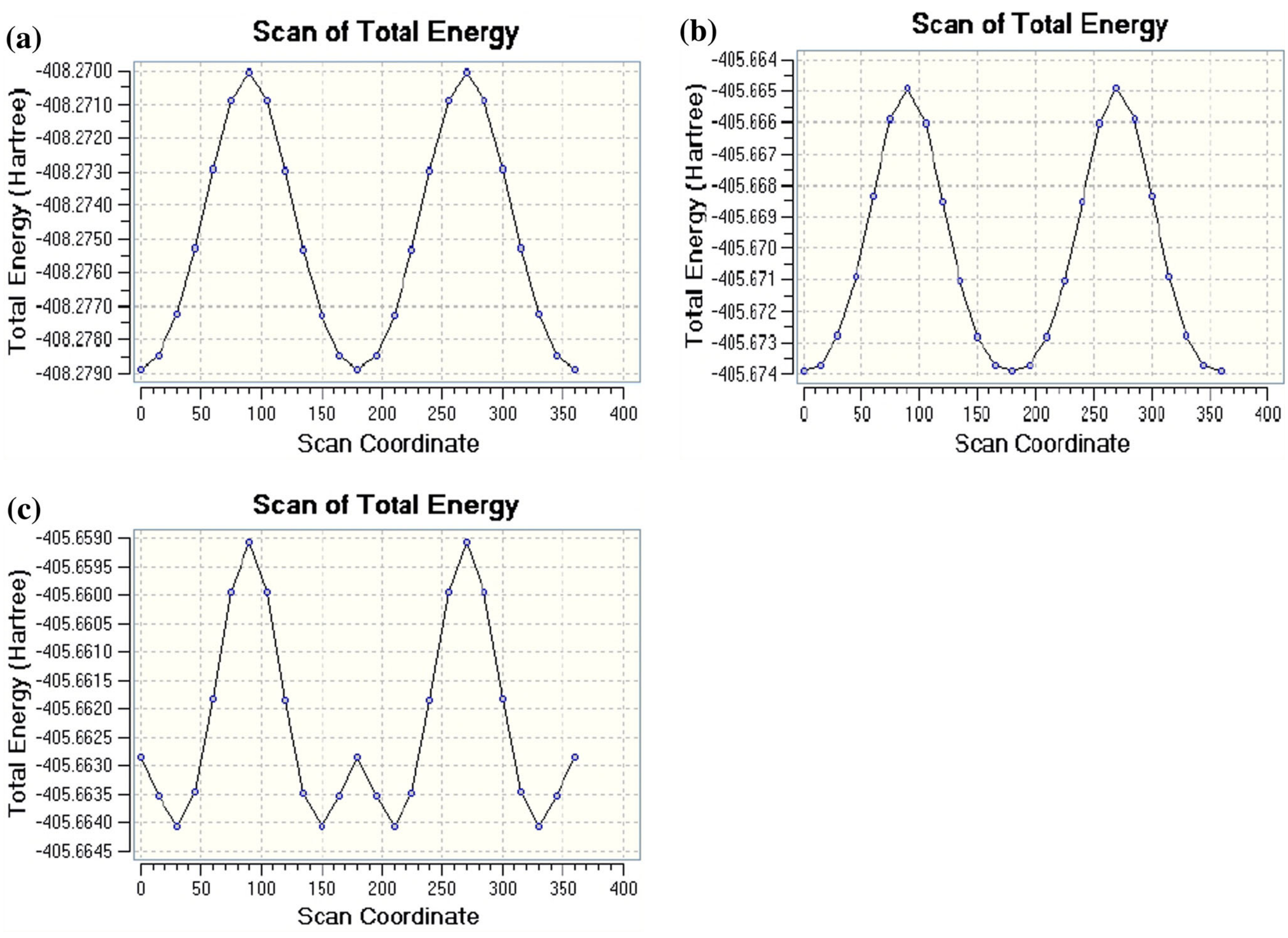


Fig. 3 Dihedral angle versus total energy plots for Compound **6**: **a** 'out–out', **b** 'in–out' and **c** 'in–in' conformations

2.3 Electrochemistry

The electrolyte solution was 1 M $LiFePO_4$ dissolved in ethylene carbonate (EC)/diethyl carbonate (DEC)/ethyl methyl carbonate (EMC) at a 1:1:1 volume ratio. The bare electrolyte was supplied by Guotai Huarong Co., Ltd., China, and used as the electrolyte without further treatment except addition of 2 % test compound. The additive was added to the bare electrolyte in an argon-filled glove box (water content less than 10 ppm). The bare EC-based electrolyte, 1.0 M $LiPF_6$ in EC/EMC/DMC (1:1:1, v/v/v), was used as control.

The cyclic voltammetry (CV) and electrochemical impedance spectroscopy (EIS) measurements were performed at 25 °C on a CHI660C electrochemical workstation (Chenhua,

Table 1 Preferred conformations and calculated energies of Compounds **1–10**

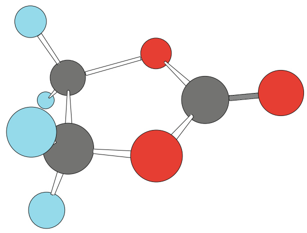
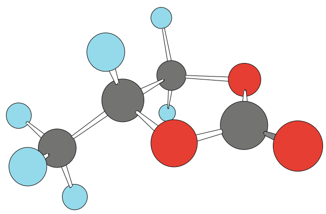
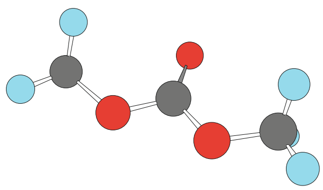
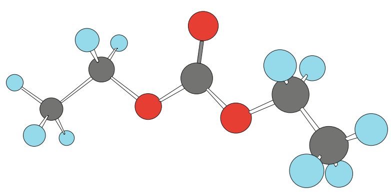
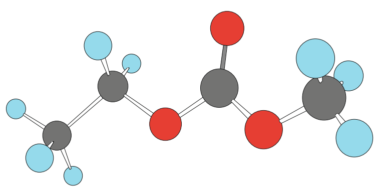
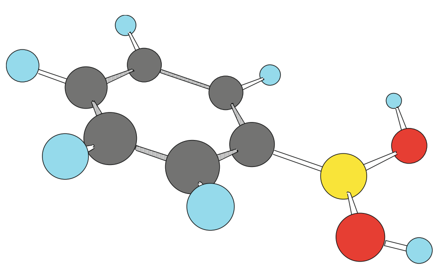
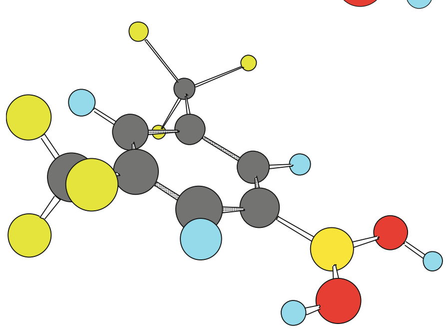
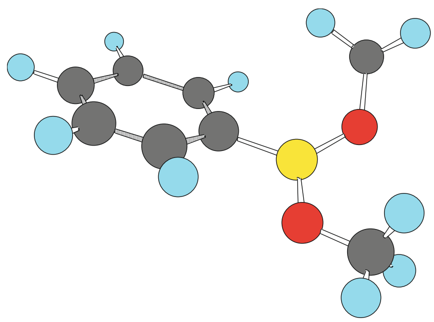
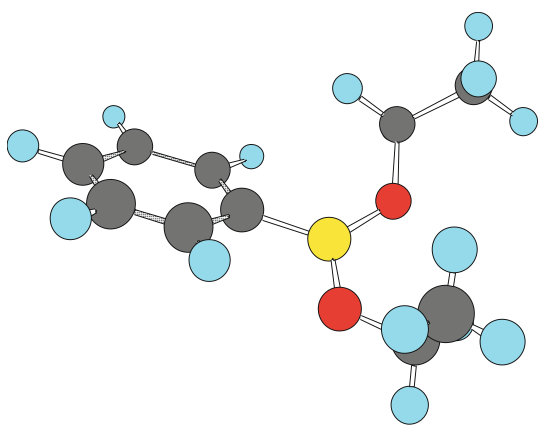
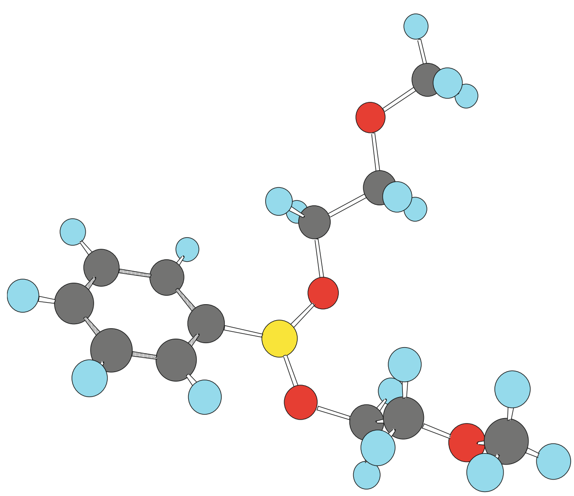
Entry	ϕ (°)	Structure	E_{HOMO} (kJ/mol)	E_{LUMO} (kJ/mol)	ΔE (kJ/mol)
1	–		−774.048	102.763	876.811
2	–		−762.735	96.462	859.197
3	–		−747.430	109.403	856.833
4	–		−734.012	28.966	762.978
5	–		−740.627	115.416	856.043
6	11.8		−651.177	−58.681	592.496
7	13.5		−743.647	−145.637	598.010

Table 1 continued

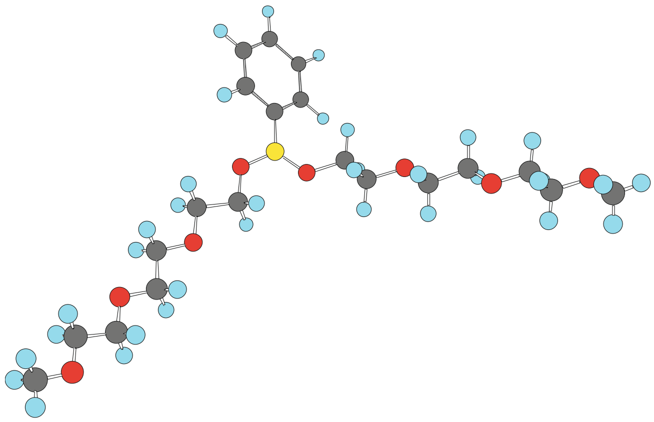
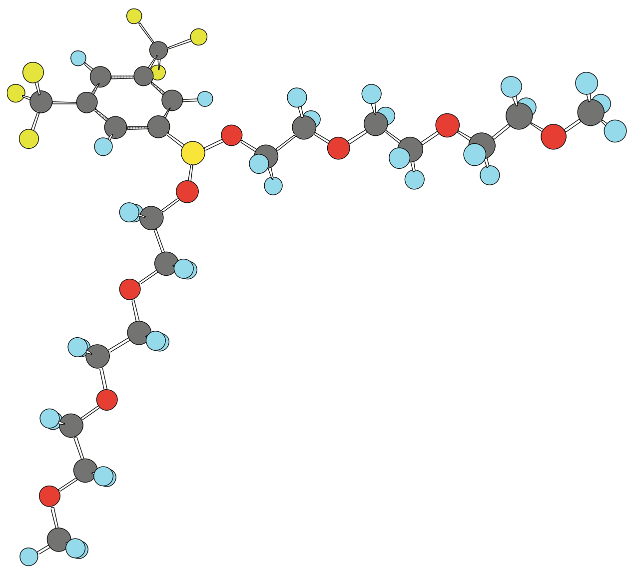
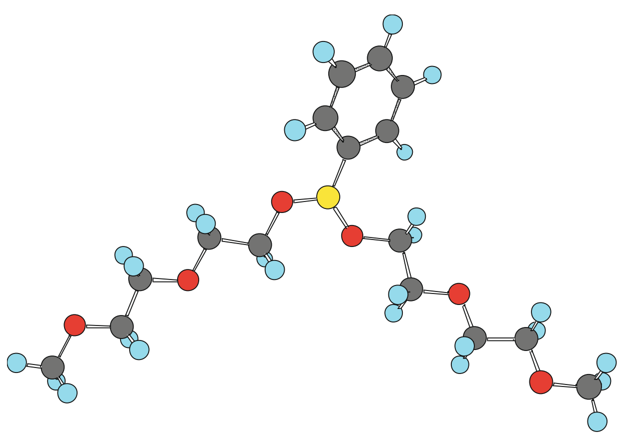
Entry	ϕ (°)	Structure	E_{HOMO} (kJ/mol)	E_{LUMO} (kJ/mol)	ΔE (kJ/mol)
8	34.9		−637.131	−39.593	597.538
9	34.5		−634.135	−36.443	597.692
10	32.0		−640.177	−46.288	539.889

China). The scan rate was 1.0 mV/s in the potential range 2.4–4.2 V.

3 Results and Discussion

Molecular orbital calculation is a very useful tool to screen additives. It is generally accepted that the structural characteristics of a molecule determine its physical and electrochemical properties. In this study, the conformations and electrochemical properties of various boronates were investigated through computational study using density functional theory (DFT) with the Becke's three-parameter hybrid method utilizing the Lee–Young–Parr correlation functional (B3LYP). After initial energy optimization using Møller–Plesset perturbation theory (MP2), the conformational preferences and energetics were investigated using DFT calculations and the 6-31G(d,p) basis set in vacuo.

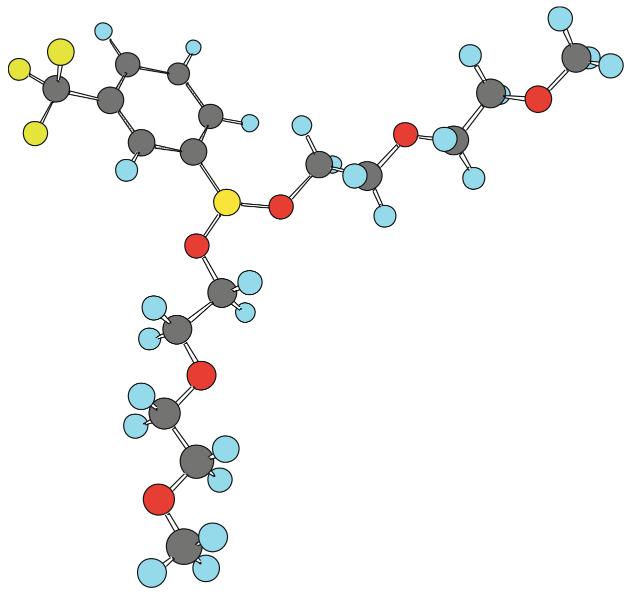
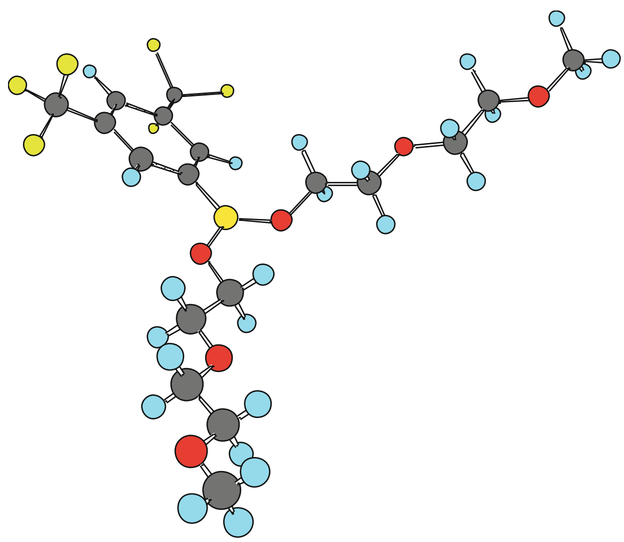
Table 1 continued

Entry	ϕ (°)	Structure	E_{HOMO} (kJ/mol)	E_{LUMO} (kJ/mol)	ΔE (kJ/mol)
11	33.6		−643.328	−48.861	594.467
12	29.5		−668.294	−132.193	536.101
13	32.4		−642.407	−48.781	593.626

3.1 Computational

Three limiting relative conformations of the boronates were considered, for convenience named 'in–in', in–out' and 'out–out' (Fig. 1). For the dihedral angle between the boronate group and the aromatic ring, 0° and 90° were taken as points of departure for the energy optimizations (Fig. 2). The conformational preference of the boronate ester groups of Compounds **6** to **15** was 'in–out' in every case, with dihedral angles between the aromatic ring and boronate group varying

Table 1 continued

Entry	ϕ (°)	Structure	E_{HOMO} (kJ/mol)	E_{LUMO} (kJ/mol)	ΔE (kJ/mol)
14	32.3		−654.457	−139.310	515.147
15	32.2		−661.599	−178.849	482.750

between 11.8° and 34.9° (Fig. 3). Calculations for Compound **16** proved impractical in terms of time and memory requirements, but this series of compounds indicates a clear trend.

The calculated LUMO energies of all boronates were significantly lower than the LUMO energies of the commonly used carbonate electrolyte solvents of lithium-ion batteries (EC, DMC and PC) and are shown in Table 1. This shows that the boronates will be easily reduced prior to the PC solvent and help the formation of the SEI film on the graphite electrode in PC-based electrolyte. This theoretical calculation suggests that these boronates will be effective electrolyte additives to a PC-based electrolyte.

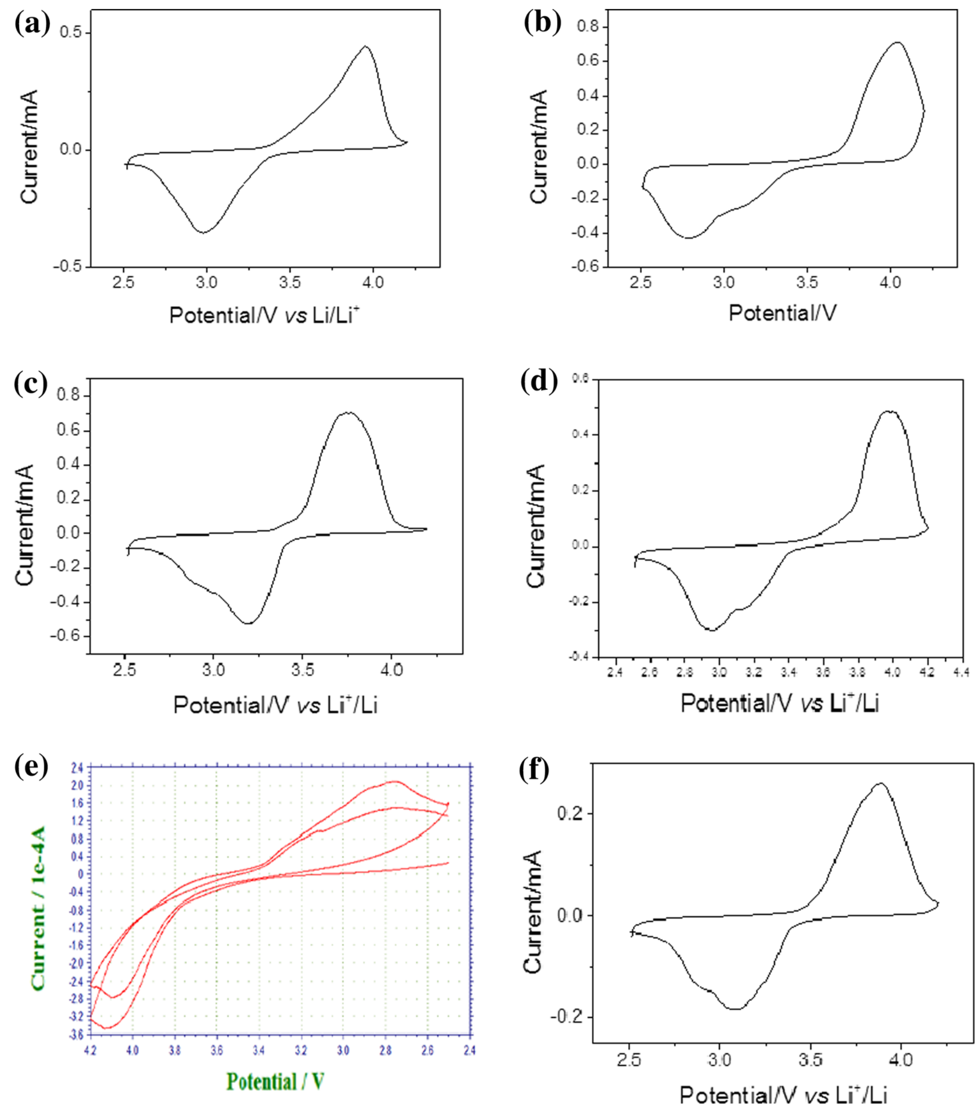
3.2 Chemical

Several EO-boronate esters (**11–16**) were synthesized in quantitative yield by condensation of suitable arylboronic acids and diethylene or triethylene glycol monomethyl ether (Scheme 2).

3.3 Electrochemical

The oxidation peak for bare electrolyte (Fig. 4) occurs at 3.80 V, and the corresponding reduction peak is located at 3.05 V. Figure 5 shows the CVs of electrolyte with 2 % w/w of each of several additives. The ionic conductivities of the

Fig. 5 Cyclic voltammograms of 1 M $LiFePO_4$ solution in EC/DMC/DEC (ethylene carbonate/dimethyl carbonate/ethyl methyl carbonate), 1/1/1, v/v/v, containing 2 % **a 11**, **b 12**, **c 13**, **d 14**, **e 15**, **f 16**



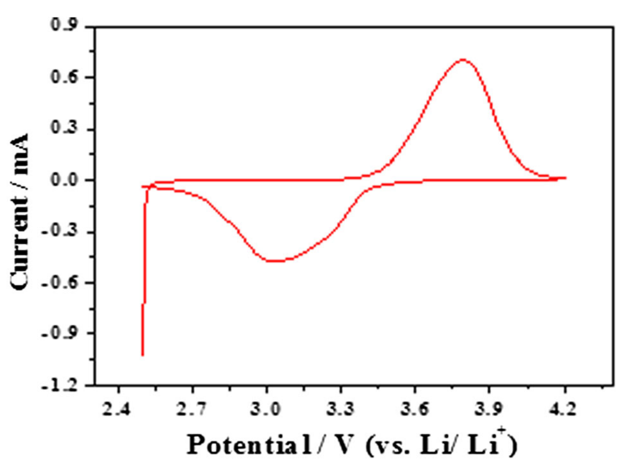


Fig. 4 Cyclic voltammogram of 1 M $LiPF_6$ solution in EC/DMC/DEC (ethylene carbonate/dimethyl carbonate/ethyl methyl carbonate), 1/1/1, v/v/v

Table 2 Ionic conductivities at 25 °C of electrolyte (1 M $LiFePO_4$ solution in EC/DMC/DEC, 1/1/1, v/v/v) with 2 % additive

Additive	Ionic conductivity (mS cm^{-1})
Bare electrolyte	0.21
11	0.26
12	0.18
13	0.24
15	0.14
16	0.17

respective solutions are listed in Table 2. Only Compounds **11** and **13** show reasonable performance, and addition to the electrolyte results in increased ionic conductivity (Table 2).

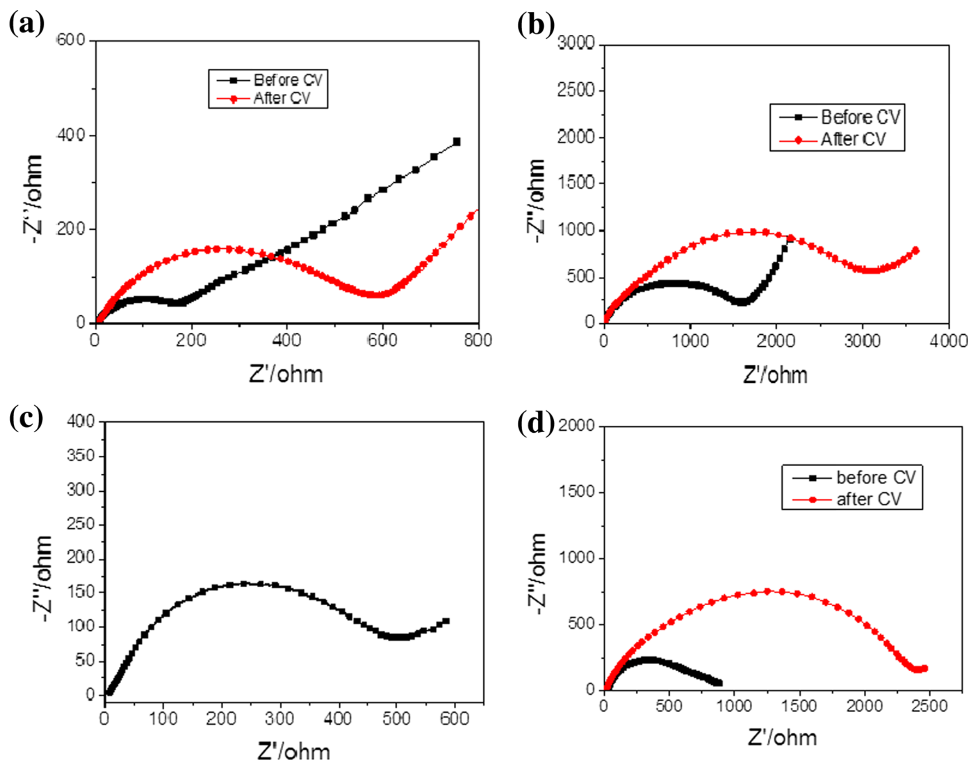


Fig. 6 EIS spectra of 1 M $LiFePO_4$ solution in EC/DMC/DEC (ethylene carbonate/dimethyl carbonate/ethyl methyl carbonate), 1/1/1, v/v/v, containing 2 % of **a 11**, **b 12**, **c 13**, **d 14**

Electrochemical impedance spectroscopy (EIS) was used for further study of the additive behaviour. Figure 6 displays the Nyquist plots. The impedance spectrum in every case is composed of a slightly depressed semicircle in the high-frequency region, which is attributed to SEI film and/or contact resistance (R_{ct}). The R_{ct} values suggest that a SEI is formed, as they are related to interfacial charged species transfer and Li^+ charge-transfer impedance. The straight line in the low-frequency Warburg region corresponds to lithium diffusion processes within the electrodes [13]. The radii of the graphs increase after CV, indicating that the additives increase the interfacial impedance. This increase is least for compounds **11** and **13**. Since excessive additive can influence the ion migration process by enlarging interfacial impedance, not more than 2 % was added in this series of experiments.

4 Conclusion

The compounds synthesized so far, especially **11** and **13**, can be expected to perform well as redox shuttles. The HOMO–LUMO energy differences indicate higher reactivity of the boronates, which would be expected to contribute to effective SEI formation. Electron-withdrawing groups (e.g. $-CF_3$) on the aromatic ring seem especially beneficial in some cases.

Acknowledgments Ms Yaqiong Yang is thanked for the CV and EIS measurements. The University of Venda and the National Research Foundation are thanked for generous financial support, and SASOL Inzalo Foundation is thanked for a fellowship for AWB.

References

1. Chen, Z.H.; Amine, K.: Functional electrolyte additive for lithium-ion batteries. Electrochem. Commun. **9**, 703–707 (2007)
2. Xiao, Q.C.; Liu, H.Y.; Xia, Q.L.; Xiao, Q.Z.; Lei, G.T.; Li, Z.H.: A nanocomposite polymer electrolyte with high-temperature stability for rechargeable lithium batteries. Arab. J. Sci. Eng. **39**, 6651–6657 (2014)
3. Moshuchak, L.M.; Bulinski, M.; Lamanna, W.M.; Wang, R.L.; Dahn, J.R.: Direct comparison of 2,5-di-*tert*-butyl-1,4-dimethoxybenzene and 4-*tert*-butyl-1,2-dimethoxybenzene as redox shuttles in $LiFePO_4$-based Li-ion cells. Electrochem. Commun. **9**, 1497–1501 (2007)
4. Zhang, S.S.: A review on electrolyte additives for lithium-ion batteries. J. Power Sources **162**, 1379–1394 (2006)
5. Xia, Q.; Wang, B.; Wu, Y.P.; Luo, H.J.; Zhao, S.Y.; van Ree, T.: Phenyl tris-2-methoxydiethoxysilane as an additive to PC-based electrolytes for lithium-ion batteries. J. Power Sources **180**, 602–606 (2008)
6. Zhang, H.P.; Xia, Q.; Wang, B.; Yang, L.C.; Wu, Y.P.; Sun, D.L.; Gan, C.L.; Luo, H.J.; Bebeda, A.W.; van Ree, T.: Vinyl-tris-(methoxydiethoxy)silane as an effective and ecofriendly flame retardant for electrolytes in lithium ion batteries. Electrochem. Comm. **11**, 526–529 (2009)
7. Li, L.L.; Li, L.; Wang, B.; Liu, L.L.; Wu, Y.P.; van Ree, T.; Thavhiwa, K.A.: Methyl phenyl bis-methoxydiethoxysilane as bi-functional additive to propylene carbonate-based electrolyte for lithium ion batteries. Electrochim. Acta **56**, 4858–4864 (2011)
8. Wang, B.; Qu, Q.T.; Xia, Q.; Wu, Y.P.; Li, X.; Gan, C.L.; van Ree, T.: Effects of 3,5-bis(trifluoromethyl)benzeneboronic

acid as an additive on electrochemical performance of propylene carbonate-based electrolytes for lithium ion batteries. Electrochim. Acta **54**, 816–820 (2008)
9. Xiao, L.F.; Ai, X.P.; Cao, Y.L.; Wang, Y.D.; Yang, H.X.: A composite polymer membrane with reversible overcharge protection mechanism for lithium ion batteries. Electrochem. Commun. **7**, 589–592 (2005)
10. Wang, R.L.; Buhrmester, C.; Dahn, J.R.: Calculations of oxidation potentials of redox shuttle additives for Li-ion cells. J. Electrochem. Soc. **153**, A445–A449 (2006)
11. Frisch, M.J.; Trucks, G.W.: Gaussian 03, Revision D.01. Gaussian, Inc., Wallingford (2004)
12. Dennington, R.; Keith, T.; Millam, J.: GaussView, Version 4.1. Semichem Inc., Shawnee Mission (2006)
13. Ufheil, J.; Baertsch, M.C.; Wursig, A.; Novak, P.: Acetone as oxidative decomposition product in propylene carbonate containing battery electrolyte. Electrochim. Acta **50**, 1733–1738 (2005)